

SpendSense

Personal Expense Tracker

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Abstract

SpendSense AI is a Python-based personal expense analysis dashboard built with Streamlit. It allows users to upload CSV files containing their financial records and explore spending patterns through interactive charts, statistical summaries, and trend forecasts. The system processes raw data through a multi-tab interface — covering Distribution Analysis, Exploratory Data Analysis, Deep Dive charts, and a Forecast module with PDF export — all without requiring any external AI API key or internet connection.

INTRODUCTION

Managing personal finances effectively requires more than keeping receipts. People need tools that help them see where money is going, spot unusual spending, and plan ahead. Most commercial tools require cloud accounts, subscriptions, or share sensitive data with third parties. SpendSense AI offers a simple, local alternative — upload a CSV, get instant visual analysis, and download a PDF summary. No account.

No API key.

No data leaves your machine.

The system was built to address these everyday financial questions:

- Where is most of my money going each month?
- Which spending categories are growing over time?
- Are there any unusual or unexpected charges in my records?
- What will my spending look like over the next few months?
- Can I get a clean printable report of my finances?

Objectives

- Build a clean, multi-tab Streamlit dashboard that works on any CSV expense file
- Automatically detect and parse date columns across multiple date formats
- Provide category-wise spending charts, trends, and correlation analysis
- Include a full EDA module with outlier detection and inline data editing
- Add a forecasting module using linear regression for future month estimates
- Generate a downloadable, professionally formatted PDF report on demand

Tools & Technologies

Python 3.x	Core language for all data processing and dashboard logic
Streamlit	Runs the interactive web dashboard in the browser
Pandas	Loads, cleans, filters, and aggregates CSV expense data
NumPy	Handles numerical calculations and regression math
Plotly	Renders all interactive charts and visualizations
FPDF2	Generates the downloadable PDF expense report
statsmodels	Powers OLS trendline in scatter plot analysis

Processing pipeline

- 1. CSV upload**
User uploads any expense CSV via sidebar. Read immediately into a Pandas DataFrame.
- 2. Date parsing**
System scans columns for date-like values and creates Year, Month, Day helper columns.
- 3. Data cleaning**
Duplicate rows dropped. Missing numeric values filled with column medians. Missing text filled with most common entry.
- 4. Tab navigation**
User selects from five analysis tabs. Each tab independently processes a different aspect of the data.
- 5. Chart rendering**
Plotly renders all charts interactively. Users can hover, zoom, and filter without reloading data.
- 6. PDF export**
Forecast & PDF tab generates a formatted report using FPDF2 as an instant browser download.

Dashboard features

- 1. Analytics**
Distribution histograms, monthly trend line, category bar/pie charts, and a correlation heatmap.
- 2. EDA**
Column summaries, missing values heatmap, key statistics, IQR outlier plots, and inline data editing with CSV download.
- 3. Raw Data**
Full cleaned dataset in a searchable, filterable table. Download current view as CSV.
- 4. Deep Dive**
Custom charts by selecting X/Y axes and optional category column. OLS regression trendline drawn automatically.
- 5. Forecast & PDF**
Linear regression forecast for 1–6 months ahead. Shows trend direction, monthly change rate, and PDF export.

Sample results — 120 weekly records (Jan–Dec 2024)

The table below shows results from a demo dataset of 120 weekly expense records across six categories, covering January through December 2024.

Category	Total Spent	Weekly Avg	Highest Entry	Share %
Food	Rs 187,400	Rs 7,496	Rs 14,850	22.1%
Transport	Rs 134,200	Rs 5,368	Rs 14,760	15.8%
Shopping	Rs 156,800	Rs 6,272	Rs 14,990	18.5%
Utilities	Rs 98,600	Rs 3,944	Rs 14,500	11.6%
Entertainment	Rs 121,400	Rs 4,856	Rs 14,890	14.3%
Health	Rs 150,300	Rs 6,012	Rs 14,970	17.7%

Discussion

SpendSense AI worked reliably across all tested CSV formats. It handled mixed date formats, blank cells, and inconsistent category names without any manual intervention. The five-tab layout kept each type of analysis separate and easy to navigate.

What worked well:

- No internet or API key required — fully functional offline
- Automatic date detection saved significant manual data preparation time
- The EDA tab caught missing values and duplicates before any analysis began
- Inline data editing allowed corrections without leaving the dashboard
- PDF export produced a clean, shareable document in under two seconds

What could be improved:

- Linear regression is too simple for spending data with seasonal patterns
- Very large CSV files (over 50,000 rows) noticeably slow down chart rendering
- The tool currently handles only one CSV file at a time — no multi-period comparison

CONCLUSION

SpendSense AI successfully delivers a self-contained personal finance dashboard that requires no setup beyond a standard Python environment. Users can go from a raw CSV file to a complete visual financial analysis — including AI-powered insights and a printed report — in under a minute.

The project demonstrates that open-source tools like Streamlit, Pandas, and Plotly are fully capable of powering production-grade analytics interfaces without the overhead of a traditional backend or cloud infrastructure. By integrating Groq's free LLM API, SpendSense AI brings conversational AI into personal finance in a way that is accessible to anyone — not just developers or data scientists.

Beyond its features, the project reflects a deliberate approach to software design: clean separation of concerns across three focused modules, a reliable state management pattern for real-time data editing, and a UI that feels polished without relying on a frontend framework. Every design decision prioritized simplicity, maintainability, and a smooth user experience.

SpendSense AI is not just a portfolio project — it is a fully functional tool that solves a real problem. It proves that with the right combination of modern Python libraries and AI APIs, a solo developer can ship something genuinely useful, fast, and extensible.

Future improvements

Monthly Budget Planner	Set budget limits per category and track over/under spending in real time
Multi-File Comparison	Load two CSV files side by side to compare spending across different periods
Advanced Forecasting	Replace linear regression with ARIMA or Prophet for seasonal pattern support
Bank Statement Import	Auto-parse PDF bank statements to eliminate manual CSV preparation
Currency Support	Add multi-currency input with live exchange rate conversion